



The relationships between CEO characteristics and strategic risk-taking in family firms

Pierluigi Martino, Alessandra Rigolini & Giuseppe D'Onza

To cite this article: Pierluigi Martino, Alessandra Rigolini & Giuseppe D'Onza (2018): The relationships between CEO characteristics and strategic risk-taking in family firms, Journal of Risk Research, DOI: [10.1080/13669877.2018.1517380](https://doi.org/10.1080/13669877.2018.1517380)

To link to this article: <https://doi.org/10.1080/13669877.2018.1517380>



Published online: 09 Oct 2018.



Submit your article to this journal [↗](#)



Article views: 19



View Crossmark data [↗](#)



The relationships between CEO characteristics and strategic risk-taking in family firms

Pierluigi Martino, Alessandra Rigolini and Giuseppe D'Onza

Department of Economic and Management, University of Pisa, Pisa, Italy

ABSTRACT

We examine the influences of chief executive officer (CEO) personal characteristics on family firms' strategic risk-taking. Building on upper echelons theory, we investigate the influences of CEO family relationships, the CEO professional education, other career experiences, tenure, and career horizon have on the risk level a company takes. By analyzing a sample of 107 Italian family firms listed on the Milan Stock Exchange, we find that company's risk-taking significantly and negatively relates to CEO family relationship and professional education, but positively to CEO career horizon. This provides support to the argument that such CEO personal characteristics are key factors in explaining differences in risk-taking among family firms. Further, our analysis of control variables shows that family firms' risk-taking relates positively to board size and negatively to company size. These results suggest that company and board characteristics also significantly influence the risk levels taken by a company.

ARTICLE HISTORY

Received 12 February 2018

Accepted 18 August 2018

KEYWORDS

Upper echelons theory; CEO characteristics; strategic risk-taking; family firms

1. Introduction

The understanding of company risk behaviors is a crucial topic, since the amount of risk a company takes is critical to its success (Damodaran 2007), and may affect organizations' investments, performance, growth, and longevity (Hiebl 2012). Among the different risk types, studies (Das and Teng 2001; Kim and Buchanan 2008) have emphasized the importance of the management of strategic risks for an organization's long-term survival and prosperity. According to a well-known definition (Baird and Thomas 1985, 231), strategic risk refers to 'corporate strategic moves that cause return to vary, that involve venturing into the unknown, and that may result in corporate ruin'. Literature on this topic has also referred to this kind of risk in terms of entrepreneurial risk or venturing risk (Zahra 2005; Naldi et al. 2007; Huybrechts, Voordeckers, and Lybaert 2013).

Several studies have analyzed the factors that influence company behaviors regarding strategic risk (Sitkin and Pablo 1992; Wiseman and Gómez-Mejía 1998; Harwood, Ward, and Chapman 2009), indicating that a company's propensity toward strategic risk-taking (or strategic risk-aversion) is the result of the combination of several factors. These include decision-maker characteristics, situational constraints, the temporal dimension, organizational characteristics, and many others (Kogan and Wallach 1967; Das and Teng 2001). Among these factors, ownership structure, particularly the presence of the largest controlling shareholder, has also been seen as

an element that can influence a firm's strategic risk-taking (Boubaker, Nguyen, and Rouatbi 2016). For instance, Faccio, Marchica, and Mura (2011) argued that firms controlled by a single under-diversified large shareholder undertake less risky investments than diversified large shareholders. In particular, when the large shareholder is a family, a firm's risk behavior may differ from that of other firms, since family firms share certain peculiarities that render them unique concerning ownership, governance, and aims (Chua, Chrisman, and Sharma 1999; Steier 2003).

To date, scholars have not reached a consensus as to whether family ownership positively or negatively affects a firm's risk-taking level. Studies built on agency theory (Naldi et al. 2007; Su and Lee 2013; Huybrechts, Voordeckers, and Lybaert 2013) claim that when there is a high concentration of wealth in one family, its extent of risk-taking is lower. Similarly, other scholars (Gómez-Mejía et al. 2007) argue that family firms generally take fewer entrepreneurial risks, since risky strategies and investments may increase the likelihood of a loss of the family's *socioemotional wealth*.

Conversely, other empirical studies (Astrachan 2003; Zahra 2003, 2005) find opposite results, highlighting that the involvement of family ownership in company management promotes risk-taking behaviors.

The mixed evidence in the literature has refocused the debate and has launched a search for other variables that may affect family firms' strategic risk-taking. Further, since the risk-taking level differs from one family firm to another, the search for new variables helps one to identify factors that can explain these differences.

We seek to contribute to this line of research by empirically examining the effects of chief executive officer (CEO) characteristics on a firm's strategic risk-taking in the context of family firms.

The decision to undertake tasks with unpredictable outcomes can be explained as a result of specific personal characteristics of decision-makers. Indeed, the personality psychology literature (Ghosh and Ray 1992; Sitkin and Weingart 1995; Highhouse and Yuce 1996) has recognized individual differences in risk-taking behaviors. Attributes such as age, gender, and personality influence each individual's risk propensity (Sitkin and Weingart 1995; Nicholson et al. 2005; Börjesson, Österberg, and Enander 2015), which leads to different risk-taking behaviors (Das and Teng 2001).

The strategic management literature has also recognized the relationship between organizational behaviors and the characteristics of a firm's leaders (Hambrick and Mason 1984). Researchers in this field have largely argued that firms' strategic choices are influenced by the cognition and values of the organization's upper echelon, and that top managers take decisions that 'fit their view of the world' (Finkelstein and Hambrick 1990, 484). Since a firm's risk-taking level is mainly the result of decisions by individuals involved in strategic decision-making processes, and their personal characteristics strongly influence the strategies' content (Datta and Guthrie 1994; Barker and Mueller 2002), the analysis of organizational leaders' tangible characteristics can help to identify factors that influence firms' risk behaviors.

Since the CEO is the formal leader of an organization and has a key role in assuring its success or failure (Finkelstein and Hambrick 1996), studies (Barker and Mueller 2002; Nelson 2005; Orens and Reheul 2013) have extensively investigated CEO characteristics, finding support for the assumption that their personal traits such as age, gender, and professional background influence organizations' outcomes. Further, in the context of family firms in which the CEO is often the company's main shareholder, they predominate among management members and dominate the process of strategy-setting, and – in turn – the firm's risk-taking level.

Nonetheless, despite CEO characteristics' influences on organizations' outcomes having been the subject of considerable research, very few studies have addressed CEO personal characteristics' influences on family firms' risk behaviors, leading to a call for more research into this topic (Gómez-Mejía, Makri, and Kintana 2010; Huybrechts, Voordeckers, and Lybaert 2013; Kraiczy, Hack, and Kellermanns 2015).

Some studies (Zahra 2005; Wang and Poutziouris 2010) have analyzed the relationship between the CEO tenure and the entrepreneurial risk-taking, providing mixed results. A more recent study by Huybrechts, Voordeckers, and Lybaert (2013) considered a CEO's family as a determinant of entrepreneurial risk-taking in family firms and found empirical support for the positive influence of a non-family CEO on a family firm's entrepreneurial risk-taking level.

Despite these studies recognizing that CEO personal characteristics are key drivers of a family firm's risk-taking, they have investigated only some CEO attributes such as tenure, age, or the CEO's family relationship (Hiebl 2012). Although in our view these attributes are key factors, these provide only a partial view of the CEO characteristics that influence a family firm's risk-taking. In our perspective, other key CEO characteristics are missing in the current debate, which must be considered. The use of upper echelons theory (Hambrick and Mason 1984) can help to solve this issue and to identify further tangible CEO characteristics that are theoretically linked to the firm's risk behavior.

We investigate the potential relationships between CEO personal characteristics and family firms' risk-taking. Specifically, drawing on upper echelons theory, we focus on the following observable managerial characteristics: (1) the CEO's belonging to the family who owns the company, (2) the CEO's educational level, referring to the CEO possession of a degree or a post-degree specialization (e.g. an MBA or a PhD), (3) career experience in other firms, (4) tenure in the company, and (5) career horizon.

To investigate these relationships, we analyzed a sample of 107 Italian family firms listed on the Milan Stock Exchange.

Our results show that the family firm's risk behavior is significantly and negatively associated with CEO family relationship and educational level, and positively with CEO career horizon. Further, analysis of the control variables shows marginally significant relationships between a family firm's risk behaviors and either its size or its board size.

This study contributes to the literature on family business and upper echelons theory. First, we contribute to the ongoing debate on risk-taking in family firms by investigating CEO personal characteristics' influences on family firms' risk behaviors. Thus, we extend the knowledge of the potential factors that influence family firms' risk-taking, responding to recent calls by Gómez-Mejía, Makri, and Kintana (2010) as well as Huybrechts, Voordeckers, and Lybaert (2013) to study CEO characteristics' influences on family firms' risk behaviors. Second, by combining insights from the literatures on corporate governance, family firms, and risk management, we provide evidence of the links between CEO characteristics and family firms' risk behaviors, extending the knowledge of individual-level factors that influence the amount of risk a firm takes. Third, this study extends the application of upper echelons theory into the family business literature: as Minichilli, Corbetta, and MacMillan (2010) note, upper echelons research has failed to consider firms with highly concentrated ownership, since it has focused particularly on large public companies. In doing our analysis in Italy, we provide evidence for a context in which ownership concentration is prevalent: indeed, Italy's equity market is characterized by a high ownership concentration level across all listed firms (Bianco and Casavola 1999; Cascino et al. 2010; Cirillo, Romano, and Pennacchio 2015). As highlighted by a recent CONSOB (Italian Oversight Board on the Milan Stock Exchange) report, in 2017, the ownership structure of Italian listed companies was highly concentrated, with an average of 47% of stakes held by the largest shareholder and almost 90% of listed companies controlled by a single shareholder or by a shareholder's agreement. Also, the report indicates that the ultimate controlling agent remains a family in most listed firms (CONSOB 2017).

Further, as Tretbar, Reimer, and Schäffer (2016) suggest, upper echelons research may benefit from being combined with the knowledge gained regarding family top management team (TMT) involvement in the business, since it has not sufficiently accounted for the influence of entrepreneurial families. Our study suggests considering the familial nature of the CEO within the upper echelons theory, since it represents a key characteristic that can influence a company's strategic

choices. The familial nature of the CEO is an observable characteristic (Hambrick and Mason 1984) that may proxy the extent of homogeneity in the dominant coalition that can affect the strategic decision-making process. In the case of family firms, the belonging of the CEO to the family owner may imply that their values are homogeneous compared to the dominant group's values.

2. Literature review and hypothesis development

2.1. Risk-taking in family firms

In the past decade, an increasing number of studies of family firms have analyzed the relationship between family firms' characteristics and firms' risk behaviors (Hiebl 2012; Huybrechts, Voordeckers, and Lybaert 2013).

Hiebl (2012) indicates that the most common argument in the literature is that family firms are more risk-averse than non-family ones. Many studies provide empirical support for this assumption. For instance, Gedajlovic, Lubatkin, and Schulze (2004) highlight that family firms tend to avoid risks and postpone risky strategic decisions as demonstrated by the lower long-term investment levels (such as the R&D spending), the lower debt to equity ratio, and the lower propensity towards innovation and internationalization compared to non-family firms.

We may argue that the literature on risk-taking in family firms has mainly been developed under three theoretical approaches. The first is based on the principles of agency theory. Building on this perspective, studies (Naldi et al. 2007; Su and Lee 2013; Huybrechts, Voordeckers, and Lybaert 2013) state that the high concentration of ownership typical of family controlled firms, in which one or more families bear the financial burden, lead to a lower company risk-taking.

The second theoretical approaches used in risk-taking of family firms research is the socioemotional wealth perspective. Following the arguments provided by this theory, scholars such as Gómez-Mejía et al. (2007) argue that family firms generally take fewer entrepreneurial risks, since risky strategies and investments may increase the likelihood of the loss of family's *socioemotional wealth*, which is the aim to reach non-financial purposes such as the ability to exercise control on the business over time, the perpetuation of family values and dynasty, and satisfying the needs for belonging and identification.

The third approach covers the studies under the stewardship perspective (Donaldson and Davis 1991), which (Zahra 2003, 2005) argue that the ownership structure of family firms ensures an alignment between the interest of the company and the family, which in turn encourages the development of new business, technologies, and entrepreneurial risk-taking. Thus, they present opposite results, highlighting that the involvement of family ownership in company management promotes risk-taking behaviors.

Thus, the question whether family firms are more or less risk-averse is still open. According to Gómez-Mejía, Makri, and Kintana (2010), the mixed results can be a result of the fact that family firms may simultaneously be risk-willing and risk-averse. These authors indicate that family firms' aim to protect their *socioemotional wealth* generally lead them to avoid high-variance investments to preserve the family's control and the company's longevity. This attitude translates into more risk-averse behavior. However, when family firms perceive a potential loss of their socioemotional wealth or of their control of the business, they are willing to accept higher risk-taking to mitigate such loss (Gómez-Mejía et al. 2007). Further, the amount of risk a family firm takes depends on the situation in which it finds itself and the de facto context in which it operates. This explanation seems consistent with the experimental psychology literature (Kahneman and Tversky 1979; Bowman 1982) that suggests that situational factors such as decision framing and the perceived decision context influence risk-taking. Indeed, according to prospect theory (Kahneman and Tversky 1979), the individual risk-taking level is fairly inconsistent across

situations; thus, a person may take risk in some circumstances, and may decide to avoid risk in other circumstances, depending on whether the choice outcome is presented as a loss or a gain (Loubergé and François Outreville 2001; Nicholson et al. 2005).

2.2. CEO influences on family firms' risk-taking

The literature has provided empirical evidence of the existence of relationships between CEO characteristics and risk behaviors. For instance, in investigating the entrepreneurial risk-taking of 209 U.S. manufacturing family firms, Zahra (2005) found CEO tenure to be negatively associated with investment in new technologies and innovation. In particular, he suggests that longer CEO tenures are conducive to strategic conformity and compliance with industry norms and practices, which in turn limit entrepreneurship. A study by Wang and Poutziouris (2010), who examined the relationships between entrepreneurs' personal features (educational level, industrial tenure, and age) and risk-taking in family firms, provides empirical support for the positive relationship between industrial tenure and risk-taking in family firms. In particular, they suggest that a family manager's higher tenure in an industry encourages its risk appetite, since the family manager has more profound knowledge of the industry, which enables them to take more risk. In addition to CEO tenure, they also found CEO age to be significantly but negatively associated to family firms' risk-taking.

Further, a recent study by Huybrechts, Voordeckers, and Lybaert (2013) examined family firms' entrepreneurial risk-taking behaviors by looking at CEOs' family relationship and CEO tenure. Based on agency theory and the socioemotional wealth perspective, they found empirical support for the argument that a non-family CEO positively influences a family firm's entrepreneurial risk-taking level, but that this effect disappears as CEO tenure increases. In particular, they argue that family CEOs discourage entrepreneurial risk-taking since they seek to preserve the family's socioemotional wealth. Thus, a non-family CEO who is not tied to the family firm may positively influence entrepreneurial risk-taking, since they have different interests and must find other ways (for instance by investing in innovation and new projects) to increase their job security. However, this positive effect decreases as CEO tenure increases, since a non-family CEO's interests tend to become increasingly aligned with those of the family owners.

As highlighted by Hiebl (2012), these studies have focused particularly on family managers' characteristics and examined only a few CEO attributes, providing mixed results. Also, they are based on agency theory and the socioemotional wealth perspective. To analyze the relationships between CEO characteristics and the risk-taking in family firms, we built on the theoretical framework of upper echelons theory (Hambrick and Mason 1984), which we describe below.

2.3. Upper echelons theory

Upper echelons theory (Hambrick and Mason 1984, 1993) argues that 'organizational outcomes, strategic choices and performance levels, are partially predicted by managerial background characteristics'. Specifically, it states that organizational outcomes and strategy may be considered as reflections of the value bases of powerful actors in organizations, especially its top managers (March and Simon 1958; Hambrick and Mason 1984).

According to this perspective, CEO experience, values, and personal characteristics influence their perceptions of the internal and external environment, which – in turn – affect their strategic decisions. Nonetheless, since values and cognitive bases are hard to measure, the authors suggest that these CEO cognitive, values, social, and psychological characteristics can be proxied by observable demographic characteristics such as age, educational background, experiences, tenure, gender, socioeconomic roots, and financial position (Hambrick and Mason 1984).

The influence of CEO demographic characteristics on organizations' outcomes has been the subject of much research. Building on the upper echelons perspective, studies have found CEO characteristics to be associated with several organizational outcomes, such as R&D spending (Barker and Mueller 2002), cash holdings (Orens and Reheul 2013), risk-taking (Farag and Mallin 2016), internationalization (Ramón-Llorens, García-Meca, and Duréndez 2017). Nonetheless, very few studies have adopted this perspective within family firms, especially in investigating their risk behaviors (Tretbar, Reimer, and Schäffer 2016). The literature has largely acknowledged the diversity of family businesses compared to other organizational forms, and their highly distinctive behaviors in areas such as internationalization, entrepreneurship, diversification, etc. owing to the involvement of one or more controlling families in the vision, ownership, management, and governance of a firm (Chua, Chrisman, and Sharma 1999; De Massis, Frattini, and Lichtenthaler 2013). These peculiarities can affect the choice of the organizational objectives to pursue (Kotlar and De Massis 2013), the acceptable risk level in strategic choices (Gómez-Mejía et al. 2007), which in turn impact on company outcomes such as the risk-taking. Further, the influence of CEO characteristics on company outcomes is particular relevant in the family business context: since the CEO is considered one of the most important and powerful organizational actors (Gómez-Mejía, Makri, and Kintana 2010; Minichilli, Corbetta, and MacMillan 2010), by having the strategic control and managing the firm (Harris and Ogbonna 2007), the CEO may have a key role in determining organizational outcomes such as the risk-taking.

Taken together, all these considerations lead us to build on upper echelons theory to analyze the links between CEO personal characteristics and family firms' strategic risk-taking.

2.3.1. The CEO family and risk-taking

According to upper echelons theory (Hambrick and Mason 1984), group heterogeneity/homogeneity is able to influence the strategic decision-making process in an organization.

Previous studies, comparing family vs. non-family firms, are based on the assumption that family firms are homogeneous (Chua et al. 2012). As Janis (1972) argues, the dominant coalition's homogeneity leads to more conservative decision-making. In particular, the homogeneity of the dominant coalition, and between the dominant coalition and the CEO, leads to the restricted generation and assessment of alternatives by restricting the growth and diversification opportunities for the firm, which in turn amounts to lower risk-taking. Thus, the higher homogeneity of the dominant group in family firms may negatively affect the risk level taken by the company.

In our view, the CEO belonging to the family owner, may be considered as an expression of the belonging of the CEO to the dominant coalition, which in family firms is represented by the family who owns the business and shares the same cognitive basis and values.

According to upper echelons theory, this means that family CEOs' decision-making processes are influenced by their cognitive basis and values, which in family business are the values of the family who owns the firm.

Studies (Gómez-Mejía et al. 2007) point out that the key reference point for families' owner decisions and behaviors is the preservation of their socioemotional wealth – the willingness to preserve family control, maintain the family's reputation and identity, entrench family executives, perpetuate the family dynasty, etc. (Sciascia et al. 2015). Thus, family CEOs are generally motivated by these non-financial, socioemotional wealth objectives (Wiseman and Gómez-Mejía 1998; Miller et al. 2014), which lead family CEOs to take fewer risks than non-family CEOs. Huybrechts, Voordeckers, and Lybaert's (2013) study provides empirical support for this argument by finding a positive influence of a non-family CEO on a family firm's entrepreneurial risk-taking level.

Thus, based on upper echelons theory and the abovementioned considerations, we propose:

Hypothesis 1: The presence of a family CEO is negatively associated with the level of family firm's risk-taking.

2.3.2. CEO educational level and risk-taking

The literature on CEO education has provided inconclusive results. For instance, a recent study by Farag and Mallin (2016) found a positive relationship between CEO education level and corporate risk-taking in China. Specifically, based on upper echelons theory, they suggest that highly educated CEOs are open to new innovative business ideas, changes, and investment opportunities and are better informed about their external environment. In turn, this facilitates the generation of entrepreneurial initiatives. Ramón-Llorens, García-Meca, and Duréndez (2017) concur, finding that the CEO education level is positively related to a family firm's ability to internationalize. However, other empirical studies found no significant relationship between CEO education level and company outcomes, such as entrepreneurial risk-taking (Wang and Poutziouris 2010), cash holdings (Orens and Reheul 2013), R&D spending (Barker and Mueller 2002), etc. For instance, a study by Barker and Mueller (2002) found a significant increase in R&D spending associated with a CEO having an undergraduate degree, but no significant relationship for higher education levels. According to them, these findings suggest that the CEO's higher education type may better predict company outcomes than amount of higher education.

CEO educational background is considered to be a key attribute that may influence top executives' cognition and values. Indeed, according to Hambrick and Mason (1984), education contributes to create person's values, knowledge, skills, and cognitive preferences, which affects executives' managerial decisions.

In particular, these authors claim that a professional education in management, i.e. the CEO has a post-degree specialization such as an MBA or a PhD, is associated with risk aversion, since such executives are primarily trained to avoid big losses and mistakes. Orens and Reheul (2013) concur by suggesting that more educated managers tend to pursue long-term development and do not invest in higher-variance projects, which can generate big losses. Consistent with upper echelons theory, we propose:

Hypothesis 2: A higher CEO educational level is negatively associated with the level of family firm's risk-taking.

2.3.3. CEO career experience in other firms and risk-taking

Upper echelons theory claims that executives' managerial decision-making is also affected by their past professional experiences. Thus, executives' past experiences represent valid proxies for their cognition, values, skills, and knowledge base as well as a valid explanation for their actions and strategic choices (Hambrick and Mason 1984; Herrmann and Datta 2006).

Particularly, Hambrick and Mason (1984) point out that experiences outside the company mean CEOs have experience that differs from those who have spent their career in one organization. This diversity of experience enables CEOs to make decisions from a broader perspective than those promoted in the company (Raskas and Brick 1992; Koellinger 2008; Zhang and Rajagopalan 2010). Farag and Mallin (2016) provide empirical support for this argument, finding a positive and highly significant relationship between CEO previous board experience and corporate risk-taking. They argue that CEOs with professional experience outside the company are more innovative and demonstrate higher openness to and awareness of new opportunities (Orens and Reheul 2013); thus, they are more risk-willing than CEOs with experience in the same company. Thus, we propose:

Hypothesis 3: CEO career experience in other firms is positively associated with the level of family firm's risk-taking.

2.3.4. CEO tenure and risk-taking

Another key managerial characteristic that can affect company outcomes is CEO tenure. In upper echelons theory, studies (Hambrick and Finkelstein 1987; Hambrick, Geletkanycz, and

Fredrickson 1993) indicate that longer CEO tenure makes companies less willing to pursue entrepreneurial initiatives. In particular, these studies argue that longer CEO tenures lead to strategic conformity and compliance with industry norms and practices, rather than upsetting the status quo via entrepreneurial activities. In turn, this reduces a firm's risk-taking propensity (Finkelstein and Hambrick 1990). Conversely, new CEOs are more open to fresh ideas, change, and experimentation. As Finkelstein and Hambrick (1996, 82) note, when CEO tenures increase, their sources of information become 'increasingly narrow and restricted, and the information is more finely filtered and distilled', and this can deprive the firm of a key source of information that can stimulate entrepreneurial activities (Zahra 2005). Zahra (2005) provides empirical support for this argument, by finding CEO tenure to be negatively related to entrepreneurial risk-taking, especially with family firms' emphasis on innovation and venturing in domestic and international markets. Empirical support for this argument has also come from a study by Farag and Mallin (2016), who found a negative relationship between CEO tenure and company risk profile. They suggest that long-tenured CEOs are less likely to consider risky decisions since they are internally focused and much less receptive to new business ideas than short-tenured CEOs.

Based on these arguments, we propose:

Hypothesis 4: Longer CEO tenure is negatively associated with the level of family firm's risk-taking.

2.3.5. CEO career horizon and company risk-taking

Several studies point out that CEO career horizon may have implications for company decision-making, and thus for company outcomes (Antia, Pantzalis, and Park 2010; Matta and Beamish 2008; McClelland, Barker, and Oh 2012; Strike et al. 2015). Specifically, the general assumption behind these studies is that CEOs with a short career horizon (i.e. near retirement), become more risk-averse; thus, they avoid risky strategies and investments with higher variance. Further, the literature on strategic risk behaviors has highlighted the temporality dimension's importance for understanding risk-taking (Das and Teng 2001). Individual risk propensity and risk behavior may differ at different points in time, which may influence the strategy planning. For this reason, although career horizon is not considered in upper echelons theory as one of the tangible characteristics of executives able to influence the decision-making process, we believe that it may strongly influence the ways executives frame their decisions about risky choices.

For instance, Matta and Beamish (2008), investigating the implications of CEO career horizon on firm risk-taking, found that a longer CEO career horizon is associated with a higher likelihood of international acquisitions. They argue that CEOs nearing retirement become more risk-averse, because they aim to conserve a legacy of success, as well as their wealth and reputation. This induces CEOs to avoid risky strategic choices that may taint their reputations in their last years of employment.

Other empirical studies support this argument, finding that older CEOs are less likely to spend money on long-term investments such as research and development (Barker and Mueller 2002) and tend to hold higher cash levels (Orens and Reheul 2013). Based on upper echelons theory, these studies claim that older CEOs are more conservative, because they may be at a point in their lives in which financial and career security are the key reference points for their choices. Further, since older CEOs have a shorter career horizon in which to recover losses, they tend to avoid risky strategies and investments in order to conserve their wealth.

Thus, based on these considerations, we posit:

Hypothesis 5: CEO career horizon is positively associated with the level of family firm's risk-taking.

3. Research method

3.1. Description of the sample

In developing our argument, we analyzed a sample of Italian family firms, listed to the Milan Stock Exchange. Italy's capital market is an ideal setting to investigate CEO demographics characteristics' influences on family firms' risk-taking, given its large number of listed family firms (Cascino et al. 2010). In Italy, family firms are a dominant organizational form (Miller et al. 2014), accounting for approximately 60% of total market capitalization (CONSOB 2013). Further, listed family-controlled firms have strong family characteristics, since controlling families are usually very involved in the company's activities, as shown by the regular appointment of family members to governance positions such as CEO (Corbetta and Minichilli 2006; Minichilli, Corbetta, and MacMillan 2010).

In line with previous studies (Anderson and Reeb 2004; Minichilli, Corbetta, and MacMillan 2010) we identified family control as the fractional equity holding by family members that allows ownership control over a company. Initially, we considered a firm in family control if one family owned more than 50% of the shares. However, given the characteristics of Italy's stock exchange, both in terms of average company size and average stock ownership (Corbetta and Minichilli 2006), consistent with previous studies of Italian listed family firms (Minichilli, Corbetta, and MacMillan 2010; Minichilli, Brogi, and Calabrò 2016), this threshold was reduced to 20%. This threshold is in line with the study by La Porta, Lopez-de-Silanes, and Shleifer (1999), who defined family firms as those in which one family owns at least a 20% share.

Following these considerations, our starting sample consisted of 161 family firms identified from the public source ORBIS (Bureau van Dijk) from 2010 to 2014. To verify whether firms in the starting sample were controlled by a family, we also compared our sample with information from CONSOB, the Italian authority for listed companies and the stock exchange market, which provides data about ownership structures and their evolution over time, checking whether, for each firm of our initial sample, there were at least two shareholders with the same last name that, together, held at least a 20% stake. From this initial sample, we excluded firms in the financial industry. Also, owing to missing data for some variables considered in our study, the final usable sample comprised 107 family firms.

We collected data on ownership, governance, and the level of family firms' risk-taking, and collected the CEO demographics characteristics for 2010 to 2014 from several sources such as annual corporate governance reports, ORBIS, LinkedIn, etc. In particular, we hand-collected information about ownership structure and corporate governance, and their evolution over time, from the annual corporate governance reports by all Italian companies. Information on CEO characteristics, including their family connection with the controlling owners, age, experience, tenure, etc. came from several sources, such as their curriculum vitae, annual corporate governance reports, company websites, LinkedIn, the Italian *who's who* list of top executives and articles from the specialized press (e.g. *Il sole 24 ore*, *Milano Finanza*). Finally, data on the companies' risk-taking levels came from ORBIS, one of the most reliable and current sources for financial information on listed companies. Table 1 highlights the sources for all variables included in the analysis.

3.2. Model specification and variable measurement description

We developed an OLS regression model to analyze the links between the CEO personal characteristics and the family firms' risk-taking. We used SPSS for our analysis, and the following model to test our research hypotheses:

$$\begin{aligned} \text{RISK-TAKING} = & \alpha + \beta_1 \text{CEO}_{\text{family}} + \beta_2 \text{CEO}_{\text{educationalLevel}} + \beta_3 \text{CEO}_{\text{otherCareerExperience}} \\ & + \beta_4 \text{CEO}_{\text{tenure}} + \beta_5 \text{CEO}_{\text{careerHorizon}} + \beta_6 \text{Board}_{\text{size}} + \beta_7 \text{Family}_{\text{directors}} \\ & + \beta_8 \text{First}_{\text{generation}} + \beta_9 \text{Family}_{\text{ownership}} + \beta_{10} \text{Firm}_{\text{size}} + \beta_{11} \text{Firm}_{\text{pastPerformance}} \\ & + \beta_{12} \text{HighTech}_{\text{industry}} + \varepsilon. \end{aligned}$$

Table 1. Sources for all variables.

Variables	Sources
RISKTAKING	Orbis
CEO_Family	Company website, Annual corporate governance reports, Italian <i>who's who</i> list of top executives, articles from the specialized press (e.g. Il sole 24 ore; Milano Finanza)
CEO_EducationaLevel	Curriculum Vitae, Company website, Italian <i>who's who</i> list of top executives, Bloomberg, LinkedIn
CEO_OtherCareerExperience	Curriculum Vitae, Company website, Italian <i>who's who</i> list of top executives, Bloomberg, LinkedIn
CEO_Tenure	Curriculum Vitae, Company website, Italian <i>who's who</i> list of top executives, Bloomberg, LinkedIn
CEO_CareerHorizon	Curriculum Vitae, Company website, Italian <i>who's who</i> list of top executives, Bloomberg, LinkedIn
Board_Size	Annual corporate governance reports
Family_Directors	Annual corporate governance reports
First_Generation	Company website, articles from the specialized press (e.g. Il sole 24 ore; Milano Finanza)
Family_Ownership	Annual corporate governance reports
Firm_Size	Orbis
Firm_PastPerformance	Orbis
HighTech_Industry	Orbis

The *dependent variable* is the family firm's risk-taking. The literature on risk conceptualizes risk as the probability distribution of returns, assuming that the larger the variance in expected returns is, the larger the risk (Markowitz 1952). Similar to previous studies (Gómez-Mejía et al. 2007; Wright et al. 2007; Huybrechts, Voordeckers, and Lybaert 2013), we used the performance variability as a proxy for the company's risk-taking, since the large variance in company performance may indicate that the firm has been pursuing risky strategies which cause greater variance in its performance outcomes. Similar to other studies (Cheng 2008; Nakano and Nguyen 2012), we measured the company's risk-taking in terms of the standard deviation of Tobin's Q, calculated as a company's market value divided by the value of its assets, using a five-year period. Further, we controlled that there was no change of CEO position during the period for which the dependent variable is calculated, so as to avoid that CEO change may influence the results.

The *explanatory variables* are CEO personal characteristics. In particular, the first explanatory variable is *the CEO's belonging to the family* (CEO_Family) who owns the company, measured as a dummy variable coded as 1 if the CEO is a member of the owning family, and 0 otherwise. As in previous studies (Miller et al. 2014), we determined a CEO's family relationship via the surname affinity with that of the controlling family. Further, we also checked the existence of other family and marriage relationships, to overcome the problem that a CEO may have a different surname to the controlling family. We identified these relationships via several sources, as reported in Table 1. The second explanatory variable is *CEO educational level* (CEO_EducationaLevel), defined as a categorical variable equal to 0 if the CEO has no degree, 1 if the CEO has a Bachelor or a Master's degree, and 2 if the CEO has a post-degree specialization (such as a PhD or an MBA), in line with Cirillo et al. (2017). Concerning *CEO career experience in other firms* (CEO_OtherCareerExperience), we used a dummy variable with a value of 1 if the CEO had other career experience as a top manager in other firms, and 0 otherwise. We measured *CEO tenure* (CEO_Tenure) using a discrete variable indicating the number of years the current CEO is in this position within the family firm, in line with previous studies (Zahra 2005; Huybrechts, Voordeckers, and Lybaert 2013). The final explanatory variable was *CEO career horizon* (CEO_CareerHorizon) measured – in line with previous studies (Matta and Beamish 2008; Strike et al. 2015) – by subtracting the CEO's current age from 70, based on the assumption that the average retirement age is 65 years, to which we added five years to capture their subsequent service on boards.

We also included several control variables in our analysis. First, since the literature points out that a company's risk-taking is influenced by its ownership and management (Jensen and Meckling 1976; Fama and Jensen 1983), we controlled for corporate governance characteristics.

Specifically, concerning board characteristics, we controlled for board size (Board_Size) in terms of the total number of directors on the board. CEOs' ability to determine a firm's strategic choice and its outcomes depends on the board's size. While studies (Nakano and Nguyen 2012; Farag and Mallin 2016) found that this variable influences a company's risk-taking, there are conflicting results in the literature. We also controlled for a family's involvement in the board (Family_Directors), measured as the percentage of family member directors on the board. A higher family involvement, owing to the presence of family members on the board, implies a greater influence of a family on a firm's strategic choices and therefore on its risk-taking (Anderson and Reeb 2004). Since a family seeks to conserve its socioemotional wealth (Gómez-Mejía et al. 2007), we predicted a negative relationship with risk-taking.

Further, we controlled for the generational stage family firms are in (Family_Generation), coded as dummy variable (with a value of 1 if a family firm is in the first generational stage, and 0 otherwise). The literature on family firms (Sciascia, Mazzola, and Chirico 2013; Kellermanns et al. 2012; Kraiczy, Hack, and Kellermanns 2015) have addressed the impacts of the controlling generation on family firms' outcomes, since family members from different generations may have different perceptions, values, financial goals, and non-financial goals, which affect their strategic decisions (Salvato, Chirico, and Sharma 2010). In particular, a controlling family's emotional attachment to a firm gradually decreases over generations. As Gómez-Mejía et al. (2011, 686) claim, 'the emphasis on preserving the family's socioemotional wealth lessens as the firm moves through generations and that financial considerations become more important as a frame of reference'. Thus, since family firms in the first generation tend to be more risk-averse, owing to the desire to conserve their socioemotional wealth, we expected a negative link between the first generation variable and a family firm's risk-taking.

Regarding ownership structure, we included in our model the percentage of the shares held by the family owner (Family_Ownership), to measure its influence on a company's risk-taking. Studies (Zahra 2005; Rigolini 2013) point out that ownership structure is a key determinant of a firm's strategic choices; thus, families who own a large percentage of shares may exercise their influence on a company's mission and strategic direction and therefore on its risk-taking. Based on the socioemotional wealth perspective (Gómez-Mejía et al. 2014), we expected a negative link between family ownership and a firm's risk-taking, since a family owner tends to avoid risky strategies and investments, to conserve its wealth, which is closely tied to the company's future.

Some studies (Bowman 1982; March and Shapira 1987) also claim that situational factors such as outcomes history, decision framing, and perceived context of situations can be key determinants of risk-taking behaviors.

Thus, since CEO decision framing is influenced by situational factors, we also controlled for a firm's past performance (Firm_PastPerformance), company size (Firm_Size), and industry (HighTech_Industry). We included company past performance, since studies (March 1988; Tversky and Kahneman 1991) suggest that in gain situation (i.e. a company shows successful past performance), the CEO tends to be more risk-averse, since they prefer sure gains and have no incentive to disrupt the status quo. Thus, we expected a negative relationship between a firm's past performance, measured as the average return on assets (ROA) over the preceding three-year period (2007–2009), and a company's risk-taking. We also controlled for company size, in terms of logarithm of the sales, since previous studies (Sathe 2003; Gómez-Mejía et al. 2007; Huybrechts, Voordeckers, and Lybaert 2013) found that this variable influences a firm's risk-taking. Finally, we controlled for industry sector by distinguishing technology sectors vs. non-technology sectors. Prior studies indicate that, in high-technology sectors, family firms are more risk-willing than family firms in non-technology sectors (Hambrick, Black, and Fredrickson 1992; Rajagopalan and Datta 1996). In line with previous studies (Kim, Pukthuanthong-Le, and Walker 2008; Cirillo, Romano, and Ardo vino 2015), we used a dummy variable, equal to 1 if the family firm belongs to the technology sector, identified by using SIC codes (283, 357, 366, 367, 318, 382, 384, 48, 737), and 0 otherwise.

4. Findings

4.1. Descriptive statistics

In this section, we provide descriptive statistics for all the variables we used. In particular, Table 2 presents descriptive statistics for the dependent and independent variables (panels A and B, respectively). Concerning the first test variable (CEO_Family), panel A shows that 42.1% of CEOs belong to the family who owns the firm. The result is consistent with previous studies on the Italian context (i.e. Minichilli et al. 2016). Regarding the second test variable (CEO_EducationaLevel), 69.2% of CEOs have a Bachelor or a Master's degree, while almost 10% have an MBA or a PhD. Also, almost 79% of the CEOs have had other career experience in other firms as a top manager and, in line with previous studies (e.g. Miller et al. 2014), the average CEO tenure is approximately seven years. Concerning the last explanatory variable (CEO_CareerHorizon), the average CEO career horizon is almost 17 years. The result is consistent with Miller et al. (2014), who found the average CEO age to be approximately 53.

Regarding the control variables, panel A indicates that 25.2% of the firms are controlled by the founders (i.e. the first generation) and 14% of the firms operate in the technology sectors (SIC codes: 283, 357, 366, 367, 318, 382, 384, 48, 737).

Table 3 presents bivariate Pearson correlations coefficient between the dependent variable RISK-TAKING and the independent variables where significant correlations are identified at 0.05 and 0.01 significance levels. It suggests that multicollinearity is not a problem in our analysis. Further, in the last column, we report the variance inflation factor (VIF) to further check for potential multicollinearity. The VIF values confirm that multicollinearity is not a concern.

To test for heteroscedasticity, we performed the Breusch–Pagan and Koenker tests. Both indicated the need to make the standard errors of our regressions robust to heteroscedasticity, which we did for all analyses.

4.2. Regression analysis

In this section, we provide the regression analysis results. Specifically, Table 4 shows the regression analysis results of Model 1, including only control variables, and the main model (Model 2),

Table 2. Descriptive statistics for the dependent and independent variables ($N = 107$).

Panel A. Discrete and dummy variables: frequency statistics			N	%
CEO_Family (H1)	Family		45	42.1%
	No family		62	57.9%
CEO_EducationaLevel (H2)	No degree		21	20.2%
	A Bachelor or a Master's degree		72	69.2%
	An MBA or a PhD		11	10.6%
CEO_OtherCareerExperience (H3)	Other career experience		82	78.8%
	No experience		22	21.2%
First_Generation	Yes		26	25.2%
	No		77	74.8%
HighTech_Industry	Yes		15	14%
	No		92	86%
Panel B. Continuous variable: descriptive statistics			Mean	Std. dev.
RISKTAKING			0.1694	0.27271
CEO_Tenure			6.7053	6.27707
CEO_CareerHorizon (H5)			17.2150	9.21138
Board_Size			9.8037	3.21975
Family_Directors			0.2676	0.15626
Family_Ownership			55.8358	14.35787
Firm_Size			8.4629	0.75633
Firm_Past_Performance			1.7702	9.47115

Table 3. Pearson correlation matrix ($N = 107$).

	RISK-TAKING	CEO_Family	CEO_Education aLevel	CEO_OtherCareer Experience	CEO_Tenure	CEO_Career Horizon	Board_Size	Family_Directors	First_Generation	Family_Ownership	Firm_Size	Firm_Past Performance	HighTech_Industry	VIF
RISK-TAKING	1													
CEO_Family	-0.106	1												1.888
CEO_EducationLevel	-0.107	-0.134	1											1.519
CEO_OtherCareer	-0.221*	-0.330**	0.171	1										1.460
Experience														
CEO_Tenure	0.052	0.348**	-0.226*	-0.265*	1									1.439
CEO_CareerHorizon	0.158	0.106	0.292**	0.004	-0.138	1								1.400
Board_Size	0.081	-0.225*	0.189	0.362**	-0.044	-0.043	1							1.641
Family_Directors	-0.076	0.573**	-0.229*	-0.113	0.261*	0.028	-0.408**	1						1.853
First_Generation	0.163	-0.005	-0.187	-0.086	-0.016	-0.218*	-0.092	-0.157	1					1.279
Family_Ownership	-0.072	0.154	-0.139	-0.069	-0.125	0.135	-0.145	0.191	-0.168	1				1.301
Firm_Size	-0.226*	-0.132	0.141	0.319**	0.003	-0.002	0.500**	-0.187	-0.199*	-0.076	1			1.666
Firm_PastPerformance	-0.168	-0.047	-0.205*	0.015	-0.181	0.084	-0.019	0.099	-0.146	0.307**	0.154	1		1.380
HighTech_Industry	0.291**	0.038	0.022	-0.189	0.081	-0.130	0.008	-0.032	0.226*	-0.143	-0.102	-0.110	1	1.160

*Significance at the 0.05 level.

**Significance at the 0.01 level.

Table 4. OLS regression analysis: dependent variable RISK-TAKING.

Regression analysis Model 1: Model with only control variables					Regression analysis Model 2: Main model				
	B	Standard error	T	Sig.		B	Standard error	T	Sig.
Board_Size	0.018	0.015	1.25	0.214	CEO_Family (H1)	-0.206	0.120	-1.72	0.012
Family_directors	-0.076	0.157	-0.48	0.632	CEO_EducationalLevel (H2)	-0.144	0.061	-2.37	0.021
First_Generation	0.032	0.045	0.71	0.478	CEO_OtherCareerExperience (H3)	-0.124	0.091	-1.37	0.177
Family_Ownership	0.000	0.001	0.25	0.806	CEO_Tenure (H4)	0.003	0.004	0.69	0.494
Firm_Size	-0.106	0.069	-1.53	0.130	CEO_CareerHorizon (H5)	0.013	0.005	2.44	0.018
Firm_PastPerformance	-0.002	0.007	-0.28	0.780	Boad_Size	0.038	0.020	1.90	0.062
HighTech_Industry	0.201	0.099	2.02	0.046	Family_directors	0.175	0.330	0.53	0.597
R ²	0.175				First_Generation	0.061	0.055	1.12	0.266
F	2.545				Family_Ownership	-0.001	0.002	-0.77	0.442
Sig. (p-value)	<.050				Firm_Size	-0.138	0.071	-1.95	0.055
N	107				Firm_PastPerformance	-0.005	0.006	-0.78	0.437
					HighTech_Industry	0.141	0.086	1.65	0.105
					R ²	0.385			
					F	3.387			
					Sig. (p-value)	<.001			
					N	107			

including all explanatory and control variables. We report each independent variable's coefficient (β) and its related significance. The last three rows provide the R^2 , the F statistics, and its p -value. Concerning the main model (Model 2), the results show that the estimated model is highly significant ($p < 0.001$), with an R^2 of 38.5%.

This model shows significant results for three of our hypotheses. Particularly, H1 is supported, indicating that a CEO's family relationship (CEO_Family) is significantly and negatively related ($p = 0.012$) to a family firm's risk-taking (RISK-TAKING). This finding suggests that family firms run by a family CEO have lower risk-taking, because a family CEO tends to avoid risky investments and strategies in order to conserve the family's socioemotional wealth. Also, the homogeneity of dominant coalition in family firms, owing to the CEO's belonging to the family, leads to restricted generation and assessment of alternatives by restricting growth and diversification opportunities for the firm, which in turn amounts to lower company risk-taking. This result is consistent with earlier studies (Gómez-Mejía et al. 2007; Huybrechts, Voordeckers, and Lybaert 2013; Miller et al. 2014) that maintain that family CEOs are more risk-averse, since they are motivated by non-financial, socioemotional wealth objectives, such as preserving family control, maintaining the family's reputation and identity, perpetuating the family dynasty, etc.

Table 4 also shows a significant and negative relationship ($p = 0.021$) between CEO educational level (CEO_EducationalLevel) and a family firm's risk-taking, which is consistent with H2. This finding, in line with upper echelons theory (Hambrick and Mason 1984), suggests that a higher CEO educational level is a key managerial characteristic in predicting company outcomes. In particular, CEOs with higher educational level (i.e. a post-degree specialization such as an MBA or a PhD) are associated with risk aversion, since they are trained primarily to avoid big losses and pursue long-term development (Orens and Reheul 2013).

Concerning our third test variable (CEO_OtherCareerExperience), although studies suggest that CEOs with other experience in other firms are more receptive to new ideas and are willing to take more risks (Hambrick and Mason 1984; Finkelstein and Hambrick 1990), we found this variable to be unrelated to the family firm's risk-taking. Thus, while this result does not support H3, it is in accordance with Serra, Trés, and Ferreira (2016), who found that CEO experience diversity is not related to a company's performance. Also, our results do not support H4: longer CEO tenure is not related to our dependent variable. While this finding is inconsistent with our expectation, it is in line with previous studies that claim that CEO tenure is a weaker predictor of company outcomes such as R&D spending (Barker and Mueller 2002) and cash holdings (Orens and Reheul 2013).

Finally, H5 is supported: we found that CEO career horizon is positively related to risk-taking ($p = 0.018$). This result, in line with previous studies (Barker and Mueller 2002; Matta and Beamish 2008), suggests that younger CEOs (i.e. CEOs with longer career horizon) are more willing to take risks than CEOs with short career horizon, because while younger CEOs have a longer career horizon in which they can recover potential losses caused by their risk-taking behaviors, older CEOs are at a point in their lives at which their reputation, financial security, and career security are important; thus, they avoid risky strategies and investments that can taint their reputation, wealth, etc.

Regarding control variables, we found significant relationships for board size (Board_Size) and company size (Firm_Size). Specifically, we found that board size positively relates to a family firm's risk-taking ($p = 0.062$). This result is consistent with Farag and Mallin (2016), who found board size to be positively and significantly related to a company's risk-taking. Further, a family firm's risk-taking is marginally significantly but negatively associated ($p = 0.055$) with company size.

4.3. Additional analysis

To test the robustness of the results in Table 4, we also performed several additional regression estimations. In particular, the literature on CEO education level suggests a reverse U-shaped

relationship (e.g. Barker and Mueller 2002). This means that CEOs with Bachelor or Master's degrees may take more risks than the CEOs without a degree. However, when CEOs are highly qualified (i.e. the CEO has a PhD or an MBA), their risk behavior change, and they seem to be more risk-averse. Thus, based on this literature, we tested for the existence of a reverse U-shaped relationship between CEO educational level and a family firm's risk-taking level.

As highlighted in Table A1 (see Appendix), the results of these additional analyses are consistent with the main model, but they don't support a reverse U-shaped relationship: the square of the variable CEO educational level is not significant. However, to further investigate the influence of CEO educational level on a firm's risk-taking, we also re-ran the main model by changing the variable CEO educational level with a series of dummy variables for various education levels, as in previous studies (Barker and Mueller 2002). The first variable, degree, takes a value of 1 if the CEO has a degree, and 0 otherwise; the second variable, post-degree specialization, takes a value of 1 if the CEO has a post-degree specialization (i.e. a PhD or an MBA), and 0 otherwise. The results of this additional analysis (reported in Table A1) show a negative relationship between education and a firm's risk-taking, both when CEOs have a degree ($p=0.043$) and a post-degree specialization ($p=0.030$). This finding supports H2, suggesting that the higher CEO educational level (i.e. a post-degree specialization such as an MBA or a PhD) is a key managerial characteristic in predicting company outcomes.

5. Conclusions, limitations, and directions for future research

In investigating the relationship between CEO characteristics and a family firms' risk-taking, we found support for H1, H2, and H5. Specifically, we found that the presence of a family CEO is significantly and negatively related to RISK-TAKING, providing support for H1. We may explain our finding arguing that, in family firms, the presence of a family CEO leads to the homogeneity of the dominant coalition, which – as Janis (1972) notes – in turn entails a restricted generation and assessment of alternatives by restricting growth and diversification opportunities for a firm, which can translated into more risk-averse behavior. Further, we maintain that the presence of family CEOs negatively affect a family firm's risk-taking, because aim to preserve the family's socio-emotional wealth; thus, it tends to avoid risky investments and strategies. Thus, a family CEO reflects the family-dominant coalition cognitive base, presenting a homogeneity of interests, aims, and aspirations that negatively affect a firm's risk-taking. This finding confirms prior studies (Wiseman and Gómez-Mejia 1998; Huybrechts, Voordeckers, and Lybaert 2013) that found family CEOs to be more risk-averse than non-family CEOs.

We also provide support for the negative relationship between CEO educational level and a firm's risk-taking. In line with upper echelons theory, this finding suggests that CEOs with higher educational level (i.e. a post degree-specialization such as an MBA or a PhD) tend to be more risk-averse and tend to pursue long-term development, since they are trained to avoid big losses and mistakes (Orens and Reheul 2013). Further, our results support H5: in particular, we found that CEO career horizon is positively associated with a family firm's risk-taking. This finding indicates that CEOs with longer career horizon (younger CEOs) are more willing to take risk. This finding supports the argument of previous studies (Matta and Beamish 2008; Farag and Mallin 2016) that claim that older CEOs (i.e. near retirement) become more conservative, since they are at a point in their lives in which the conservation of their (or their family's) financial and career security become the key reference points for their decisions.

In contrast, our findings do not support the hypotheses regarding CEO other career experience in other firms (H3) and CEO tenure (H4). Further, the results of the control variable indicate that family firms' risk-taking is positively influenced by board characteristics (board size) and negatively by company characteristics (company size), providing support to prior study findings (Sathe 2003; Huybrechts, Voordeckers, and Lybaert 2013; Farag and Mallin 2016).

Our study results have both theoretical and practical implications. First, this study contributes to the existing literature on family firms' risk-taking. Specifically, since the literature focused mostly on the classical duality of ownership structure and company risk-taking, by investigating CEO characteristics' influences, we have extended the knowledge about the potential factors that influence family firms' risk-taking, responding to calls by Gómez-Mejía, Makri, and Kintana (2010), as well as Huybrechts, Voordeckers, and Lybaert (2013) for more research into CEO characteristics' influences on family firms' risk-taking. Indeed, despite the influence of CEO demographic characteristics on organizations' outcomes having been the subject of much research, very few studies have addressed CEO personal characteristics' influences have on family firms' risk-taking. Also, previous studies have investigated only a part of CEO attributes, providing conflicting results (Hiebl 2012). Second, we have extended the application of upper echelons perspective into the family business literature, since previous research has focused particularly on large public companies (Minichilli, Corbetta, and MacMillan 2010), and very few studies have adopted this perspective within family firms, especially in investigating their risk behaviors (Tretbar, Reimer, and Schäffer 2016). Our study findings provide empirical support for the argument that organizational outcomes can be predicted by managerial background characteristics (Hambrick and Mason 1984). Particularly, our findings indicate that some CEO characteristics – i.e. the CEO belonging to the family owner, CEO educational level, and CEO career horizon are key factors that determine CEOs' risk-taking behaviors, which in turn affect firms' risk-taking. Thus, in line with upper echelons theory, we suggest that these CEO characteristics may help to explain differences in risk-taking among family firms. This applies particularly to family-controlled firms in which the CEO is the most powerful organizational actor, and strongly influences a company's strategic decision-making process and therefore company outcomes (Minichilli, Corbetta, and MacMillan 2010; Miller et al. 2014). Further, the empirical evidence of the influence of CEO type (family vs. no family) on risk-taking suggests considering this variable as observable managerial characteristic within the upper echelon perspective when applied to the context of family firms. Upper echelon theory suggests that group heterogeneity/homogeneity may influence the strategic decision-making process in an organization. In the context of family firms, the CEO belonging to the family owner may be considered an expression of the CEO belonging to the dominant coalition which, in family firms, is represented by the family who owns the business and shares the same cognitive basis and values. This means that the dominant coalition and the CEO, in the context of family firms, may be characterized by greater homogeneity within the group. Thus, the CEO's familial nature may help one to understand the extent of group homogeneity (Hambrick and Mason 1984) and how it may affect the strategic decision-making process and company strategic risk behaviors.

This study also has practical implications. First, companies may benefit from our research findings, since we provide evidence of variables that significantly affect family firms' risk-taking. For instance, internal and external stakeholders interested in knowing a company's risk-taking may benefit from our research in evaluating the potential factors that influence this crucial component of a risk management system, and can use our findings to support their analyses. Our study findings may also be used to support CEO selection processes, since it helps to identify some personal traits that should be considered when hiring or promoting someone into the CEO position. For instance, family firms who wish to pursue risky strategies and investments (e.g. innovation, internationalization, etc.) when hiring a CEO or promoting someone into the CEO position, should consider appointing a non-family and younger CEO so as to promote entrepreneurial activities.

Our study results should be interpreted with caution, owing to certain limitations. First, in investigating a CEO's influence on a family firm's risk-taking, we may have missed other CEO personal characteristics that may be included in the upper echelons framework (such as the CEO financial position and functional tracks), owing to the absence of publicly available information.

Future research can add to our variables so as to analyze the relationships between other CEO characteristics and family firms' risk-taking.

Further, since our study is limited to a fairly small sample and to one country (Italy), any generalization of the results should be done with caution. Future research could explore whether our findings can be generalized to other countries.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

- Anderson, R. C., and D. M. Reeb. 2004. "Board Composition: Balancing Family Influence in S&P 500 Firms." *Administrative Science Quarterly* 49 (2): 209–237.
- Antia, M., C. Pantzalis, and J. C. Park. 2010. "CEO Decision Horizon and Firm Performance: An Empirical Investigation." *Journal of Corporate Finance* 16: 288–301.
- Astrachan, J. H. 2003. "Commentary on the Special Issue: The Emergence of a Field." *Journal of Business Venturing* 18 (5): 567–573.
- Baird, I. S., and H. Thomas. 1985. "Toward a Contingency Model of Strategic Risk Taking." *Academy of Management Review* 10: 230–243.
- Barker III, V. L., and G. C. Mueller. 2002. "CEO Characteristics and Firm R&D Spending." *Management Science* 48 (6): 782–801.
- Bianco, M., and P. Casavola. 1999. "Italian Corporate Governance: Effects On Financial Structure and Firm Performance." *European Economic Review* 43 (4–6): 1057–1069.
- Börjesson, M., J. Österberg, and A. Enander. 2015. "Risk Propensity within the Military: A Study of Swedish Officers and Soldiers." *Journal of Risk Research* 18 (1): 55–68. doi:10.1080/13669877.2013.879489.
- Boubaker, S., P. Nguyen, and W. Rouatbi. 2016. "Multiple Large Shareholders and Corporate Risk-Taking: Evidence from French Family Firms." *European Financial Management* 22 (4): 697–745.
- Bowman, E. H. 1982. "Risk Seeking by Troubled Firms." *Sloan Management Review* 23 (4): 33.
- Cascino, S., A. Pugliese, D. Mussolino, and C. Sansone. 2010. "The Influence of Family Ownership on the Quality of Accounting Information." *Family Business Review* 23 (3): 246–265.
- Cheng, S. 2008. "Board Size and the Variability of Corporate Performance." *Journal of Financial Economics* 87: 157–176.
- Chua, J. H., J. J. Chrisman, and P. Sharma. 1999. "Defining the Family Business by Behavior." *Entrepreneurship Theory and Practice* 23 (4): 19–39.
- Chua, J. H., J. J. Chrisman, L. P. Steier, and S. B. Rau. 2012. "Sources of Heterogeneity in Family Firms: An Introduction." *Entrepreneurship Theory and Practice* 36 (6): 1103–1113.
- Cirillo, A., D. Mussolino, M. Romano, and R. Viganò. 2017. "A Complicated Relationship: Family Involvement in the Top Management Team and Post-IPO Survival." *Journal of Family Business Strategy* 8 (1): 42–56.
- Cirillo, A., M. Romano, and O. Ardovino. 2015. "Does Family Involvement Foster IPO Value? Empirical Analysis on Italian Stock Market." *Management Decision* 53 (5): 1125–1154.
- Cirillo, A., M. Romano, and L. Pennacchio. 2015. "All the Power in Two Hands: The Role of CEOs in Family IPOs." *European Management Journal* 33 (5): 392–406.
- CONSOB (Commissione Nazionale per le Società e la Borsa). 2013. Report on corporate governance of Italian listed companies.
- CONSOB (Commissione Nazionale per le Società e la Borsa). 2017. Report on corporate governance of Italian listed companies.
- Corbetta, G., and A. Minichilli. 2006. "Board of Directors in Italian Public Family Controlled Companies'." In *Family Business Research Handbook*, edited by P. Poutziouris, K. Smyrniotis, and S. Klein, 488–500. Northampton: Elgar.
- Damodaran, A. 2007. *Strategic Risk Taking: A Framework for Risk Management*. Philadelphia: Wharton School Publishing.
- Das, T. K., and B. S. Teng. 2001. "Strategic Risk Behaviour and Its Temporalities: Between Risk Propensity and Decision Context." *Journal of Management Studies* 38 (4): 515–534. doi:10.1111/1467-6486.00247.
- Datta, D. K., and J. P. Guthrie. 1994. "Executive Succession: Organizational Antecedents of CEO Characteristics." *Strategic Management Journal* 15 (7): 569–577.
- De Massis, A., F. Frattini, and U. Lichtenthaler. 2013. "Research on Technological Innovation in Family Firms: Present Debates and Future Directions." *Family Business Review* 26 (1): 10–31.

- Donaldson, L., and J. H. Davis. 1991. "Stewardship Theory or Agency Theory: CEO Governance and Shareholder Returns." *Australian Journal of Management* 16 (1): 49–64.
- Faccio, M., M. T. Marchica, and R. Mura. 2011. "Large Shareholder Diversification and Corporate Risk-Taking." *Review of Financial Studies* 24 (11): 3601–3641.
- Fama, E. F., and M. C. Jensen. 1983. "Separation of Ownership and Control." *Journal of Law and Economics* 26 (2): 301–325.
- Farag, H., and C. Mallin. 2016. "The Influence of CEO Demographic Characteristics on Corporate Risk-Taking: Evidence from Chinese IPOs." *The European Journal of Finance* 24 (16): 1528–1551.
- Finkelstein, S., and D. C. Hambrick. 1990. "Top-Management-Team Tenure and Organizational Outcomes: The Moderating Role of Managerial Discretion." *Administrative Science Quarterly* 35 (3): 484–503.
- Finkelstein, S., and D. C. Hambrick. 1996. *Strategic Leadership: Top Executives and Their Effects on Organizations*. Minneapolis, St Paul: West Publishing Company.
- Gedajlovic, E., M. H. Lubatkin, and W. S. Schulze. 2004. "Crossing the Threshold from Founder Management to Professional Management: A Governance Perspective." *Journal of Management Studies* 41 (5): 899–912.
- Ghosh, D., and M. R. Ray. 1992. "Risk Attitude, Ambiguity Intolerance and Decision Making: An Exploratory Investigation." *Decision Sciences* 23 (2): 431–444.
- Gómez-Mejía, L. R., J. T. Campbell, G. Martin, R. E. Hoskisson, M. Makri, and D. G. Sirmon. 2014. "Socioemotional Wealth as a Mixed Gamble: Revisiting Family Firm R&D Investments with the Behavioral Agency Model." *Entrepreneurship Theory and Practice* 52 (6): 1351–1374.
- Gómez-Mejía, L. R., C. Cruz, P. Berrone, and J. De Castro. 2011. "The Bind That Ties: Socioemotional Wealth Preservation in Family Firms." *Academy of Management Annals* 5 (1): 653–707.
- Gómez-Mejía, L. R., M. Makri, and M. L. Kintana. 2010. "Diversification Decisions in Family-Controlled Firms." *Journal of Management Studies* 47 (2): 223–252.
- Gómez-Mejía, L. R., K. Takács Haynes, M. Núñez-Nickel, K. J. L. Jacobson, and J. Moyano-Fuentes. 2007. "Socioemotional Wealth and Business Risks in Family-Controlled Firms: Evidence from Spanish Olive Oil Mills." *Administrative Science Quarterly* 52 (1): 106–137.
- Hambrick, D. C., S. Black, and J. W. Fredrickson. 1992. "Executive Leadership of the High Technology Firm: What is Special About It." *Advances in Global High-Technology Management* 2: 3–18.
- Hambrick, D. C., and S. Finkelstein. 1987. "Managerial Discretion: A Bridge between Polar Views of Organizational Outcomes." *Research in Organizational Behavior* 9: 369–406.
- Hambrick, D. C., M. A. Geletkanycz, and J. W. Fredrickson. 1993. "Top Executive Commitment to the Status Quo: Some Tests of Its Determinants." *Strategic Management Journal* 14 (6): 401–418.
- Hambrick, D. C., and P. A. Mason. 1984. "Upper Echelons: The Organization as a Reflection of Its Top Managers." *Academy of Management Review* 9 (2): 193–206.
- Harris, L. C., and E. Ogbonna. 2007. "Ownership and Control in Closely-Held Family-Owned Firms: An Exploration of Strategic and Operational Control." *British Journal of Management* 18 (1): 5–26.
- Harwood, I. A., S. C. Ward, and C. B. Chapman. 2009. "A Grounded Exploration of Organisational Risk Propensity." *Journal of Risk Research* 12 (5): 563–579. doi:[10.1080/13669870802497751](https://doi.org/10.1080/13669870802497751).
- Herrmann, P., and D. K. Datta. 2006. "CEO Experiences: Effects on the Choice of FDI Entry Mode." *Journal of Management Studies* 43 (4): 755–778.
- Hiebl, M. R. W. 2012. "Risk Aversion in Family Firms: what Do we Really Know?" *The Journal of Risk Finance* 14 (1): 49–70.
- Highhouse, S., and P. Yuce. 1996. "Perspectives, Perceptions and Risk-Taking Behaviour." *Organizational Behavior and Human Decision Processes* 65 (2): 159–167.
- Huybrechts, J., W. Voordeckers, and N. Lybaert. 2013. "Entrepreneurial Risk Taking of Private Firms: The Influence of a Nonfamily CEO and the Moderating Effect of CEO Tenure." *Family Business Review* 26 (2): 161–179.
- Janis, I. L. 1972. *Victims of Groupthink*. Boston: Houghton Mifflin.
- Jensen, M. C., and W. H. Meckling. 1976. "Theory of the Firm: Managerial Behaviour, Agency Costs and Ownership Structure." *Journal of Financial Economics* 3 (4): 305–360.
- Kahneman, D., and A. Tversky. 1979. "Prospect Theory: An Analysis of Decisions under Risk." *Econometrica* 47 (2): 263–291.
- Kellermanns, F. W., K. A. Eddleston, R. Sarathy, and F. Murphy. 2012. "Innovativeness in Family Firms: A Family Influence Perspective." *Small Business Economics* 38 (1): 85–101. doi:[10.1007/s11187-010-9268-5](https://doi.org/10.1007/s11187-010-9268-5).
- Kim, J., K. Pukthuanthong-Le, and T. Walker. 2008. "Leverage and IPO under-Pricing: High-Tech versus Low-Tech IPOs." *Management Decision* 46 (1): 106–130.
- Kim, K., and R. Buchanan. 2008. "CEO Duality Leadership and Firm Risk-Taking Propensity." *Journal of Applied Business Research* 24 (1): 27.
- Koellinger, P. D. 2008. "Why Are Some Entrepreneurs More Innovative than Others?" *Small Business Economics* 31 (1): 21–37.
- Kogan, N., and M. A. Wallach. 1967. "Risk Taking as a Function of the Situation, the Person, and the Group." *New Directions in Psychology* 3: 111–278.

- Kotlar, J., and A. De Massis. 2013. "Goal Setting in Family Firms: Goal Diversity, Social Interactions, and Collective Commitment to Family-Centered Goals." *Entrepreneurship Theory and Practice* 37 (6): 1263–1288.
- Kraczy, N. D., A. Hack, and F. W. Kellermanns. 2015. "What Makes a Family Firm Innovative? CEO Risk-Taking Propensity and the Organizational Context of Family Firms." *Journal of Product Innovation Management* 32 (3): 334–348. doi:[10.1111/jpim.12203](https://doi.org/10.1111/jpim.12203).
- La Porta, R., F. Lopez-de-Silanes, and A. Shleifer. 1999. "Corporate Ownership around the World." *Journal of Finance* 54 (2): 471–517.
- Loubergé, H., and J. François Outreville. 2001. "Risk Taking in the Domain of Losses: Experiments in Several Countries." *Journal of Risk Research* 4 (3): 227–236. doi:[10.1080/13669870010016930](https://doi.org/10.1080/13669870010016930).
- March, J. G. 1988. "Variable Risk Preferences and Adaptive Aspirations." *Journal of Economic Behavior & Organization* 9 (1): 5–24.
- March, J. G., and H. A. Simon. 1958. Organizations.
- March, J. G., and Z. Shapira. 1987. "Managerial Perspectives on Risk and Risk Taking." *Management Science* 33 (11): 1404–1418.
- Markowitz, H. 1952. "Portfolio Selection." *The Journal of Finance* 7 (1): 77–91.
- Matta, E., and P. W. Beamish. 2008. "The Accentuated CEO Career Horizon Problem: Evidence from International Acquisitions." *Strategic Management Journal* 29 (7): 683–700.
- McClelland, P. L., V. L. Barker III, and W. Y. Oh. 2012. "CEO Career Horizon and Tenure: Future Performance Implications under Different Contingencies." *Journal of Business Research* 65: 1387–1393.
- Miller, D., I. Le Breton-Miller, A. Minichilli, G. Corbetta, and D. Pittino. 2014. "When Do Non-Family CEOs Outperform in Family Firms? Agency and Behavioural Agency Perspectives." *Journal of Management Studies* 51 (4): 547–572.
- Minichilli, A., M. Brogi, and A. Calabrò. 2016. "Weathering the Storm: Family Ownership, Governance, and Performance through the Financial and Economic Crisis." *Corporate Governance: An International Review* 24 (6): 552–568.
- Minichilli, A., G. Corbetta, and I. C. MacMillan. 2010. "Top Management Team in Family Controlled Companies: familiness, Faultiness, and the Impact on Financial Performance." *Journal of Management Studies* 47 (2): 205–222.
- Nakano, M., and P. Nguyen. 2012. "Board Size and Corporate Risk Taking: Further Evidence from Japan." *Corporate Governance: An International Review* 20 (4): 369–387.
- Naldi, L., M. Nordqvist, K. Sjöberg, and J. Wiklund. 2007. "Entrepreneurial Orientation, Risk Taking, and Performance in Family Firms." *Family Business Review* 20 (1): 33–47.
- Nelson, J. 2005. "Corporate Governance Practices, CEO Characteristics and Firm Performance." *Journal of Corporate Finance* 11 (1–2): 197–228.
- Nicholson, N., E. Soane, M. Fenton-O'Creevy, and P. Willman. 2005. "Personality and Domain-Specific Risk Taking." *Journal of Risk Research* 8 (2): 157–176.
- Orens, R., and A. M. Reheul. 2013. "Do CEO Demographics Explain Cash Holdings in SMEs?" *European Management Journal* 31 (6): 549–563.
- Rajagopalan, N., and D. Datta. 1996. "CEO Characteristics: Does Industry Matter?" *Academy of Management Journal* 39 (1): 197–215.
- Ramón-Llorens, M. C., E. García-Meca, and A. Duréndez. 2017. "Influence of CEO Characteristics in Family Firms Internationalization." *International Business Review* 26 (4): 786–799.
- Raskas, D. F., and D. C. Brick. 1992. "Multifunctional Managerial Development: A Framework for Evaluating the Options." *Organizational Dynamics* 21 (2): 5–17.
- Rigolini, A. 2013. *Corporate Entrepreneurship and Corporate Governance*. Milano: McGraw-Hill.
- Salvato, C., F. Chirico, and P. Sharma. 2010. "A Farewell to the Business: Championing Exit and Continuity in Entrepreneurial Family Firms." *Entrepreneurship and Regional Development* 22 (3–4): 321–348.
- Sathe, V. 2003. *Corporate Entrepreneurship: Top Managers and New Business Creation*. Cambridge, England: Cambridge University Press.
- Sciascia, S., P. Mazzola, and F. Chirico. 2013. "Generational Involvement in the Top Management Team of Family Firms: Exploring Nonlinear Effects on Entrepreneurial Orientation." *Entrepreneurship Theory and Practice* 37 (1): 69–85.
- Sciascia, S., M. Nordqvist, P. Mazzola, and A. De Massis. 2015. "Family Ownership and R&D Intensity in Small- and Medium-Sized Firms." *Journal of Product Innovation Management* 32 (3): 349–360.
- Serra, F. R., G. Trê, and M. P. Ferreira. 2016. "The CEO' Effect on the Performance of Brazilian Companies: An Empirical Study Using Measurable Characteristics." *European Management Review* 13 (3): 193–205.
- Sitkin, S. B., and A. L. Pablo. 1992. "Reconceptualizing the Determinants of Risk Behavior." *Academy of Management Review* 17 (1): 9–38.
- Sitkin, S. B., and L. R. Weingart. 1995. "Determinants of Risky Decision-Making Behavior: A Test of the Mediating Role of Risk Perceptions and Propensity." *Academy of Management Journal* 38 (6): 1573–1592.
- Steier, L. 2003. "Variants of Agency Contracts in Family Financed Ventures as a Continuum of Familial Altruistic and Market Rationalities." *Journal of Business Venturing* 18 (5): 597–618.

- Strike, V. M., P. Berrone, S. G. Sapp, and L. Congiu. 2015. "A Socioemotional Wealth Approach to CEO Career Horizons in Family Firms." *Journal of Management Studies* 52 (4): 555–583.
- Su, W., and C. Y. Lee. 2013. "Effects of Corporate Governance on Risk Taking in Taiwanese Family Firms during Institutional Reform." *Asia Pacific Journal of Management* 30 (3): 809–828.
- Tretbar, T., M. Reimer, and U. Schäffer. 2016. *Upper Echelons in Family Firms: What We Know and Still Can Learn about Family-TMT-Involvement*, edited by F. Kellermanns and F. Hoy, 90–117. Routledge, London: The Routledge Companion to Family Business.
- Tversky, A., and D. Kahneman. 1991. "Loss Aversion in Riskless Choice: A Reference-Dependent Model." *The Quarterly Journal of Economics* 106 (4): 1039–1061.
- Wang, Y., and P. Poutziouris. 2010. "Entrepreneurial Risk Taking: Empirical Evidence from UK Family Firms." *International Journal of Entrepreneurial Behavior & Research* 16 (5): 370–388.
- Wiseman, R. M., and L. R. Gómez-Mejía. 1998. "A Behavioral Agency Model of Managerial Risk Taking." *Academy of Management Review* 23 (1): 133–153.
- Wright, P., M. Kroll, J. A. Krug, and M. Pettus. 2007. "Influences of Top Management Team Incentives on Firm Risk Taking." *Strategic Management Journal* 28 (1): 81–89. doi:[10.1002/smj.548](https://doi.org/10.1002/smj.548).
- Zahra, S. A. 2003. "International Expansion of U.S. manufacturing Family Businesses: The Effect of Ownership and Involvement." *Journal of Business Venturing* 18 (4): 495–512.
- Zahra, S. A. 2005. "Entrepreneurial Risk Taking in Family Firms." *Family Business Review* 18 (1): 23–40. doi:[10.1111/j.1741-6248.2005.00028](https://doi.org/10.1111/j.1741-6248.2005.00028).
- Zhang, Y., and N. Rajagopalan. 2010. "Once an Outsider, Always an Outsider? CEO Origin, Strategic Change, and Firm Performance." *Strategic Management Journal* 31 (3): 334–346.

Appendix

Table A1. Additional analyses: CEO educational level tests for the reverse U-shaped relationship.

Additional analysis 1: (CEO educational level square transformation)					Additional analysis 2: (CEO educational level with dummy variables)				
	<i>B</i>	Standard error	<i>T</i>	Sig.		<i>B</i>	Standard error	<i>T</i>	Sig.
CEO_Family (H1)	−0.206	0.122	−1.69	0.014	CEO_Family (H1)	−0.206	0.122	−1.69	0.014
CEO_EducationalLevel (H2)	−0.152	0.115	−1.33	0.189	Degree_Education	−0.148	0.071	−2.07	0.043
CEO_OtherCareerExperience (H3)	−0.125	0.091	−1.37	0.177	Post-degree_Specialization	−0.286	0.129	−2.22	0.030
CEO_Tenure (H4)	0.003	0.004	0.71	0.479	CEO_OtherCareer Experience (H3)	−0.125	0.091	−1.37	0.177
CEO_CareerHorizon (H5)	0.013	0.005	2.49	0.016	CEO_Tenure (H4)	0.003	0.004	0.71	0.479
Boad_Size	0.038	0.020	1.91	0.061	CEO_CareerHorizon (H5)	0.013	0.005	2.49	0.016
Family_directors	0.173	0.341	0.51	0.613	Boad_Size	0.038	0.020	1.91	0.061
First_Generation	0.061	0.055	1.10	0.274	Family_directors	0.173	0.341	0.51	0.613
Family_Ownership	−0.001	0.002	−0.78	0.440	First_Generation	0.061	0.055	1.10	0.274
Firm_Size	−0.138	0.072	−1.91	0.060	Family_Ownership	−0.001	0.002	−0.78	0.440
Firm_PastPerformance	−0.005	0.006	−0.77	0.446	Firm_Size	−0.138	0.072	−1.91	0.060
HighTech_Industry	0.140	0.090	1.56	0.123	Firm_PastPerformance	−0.005	0.006	−0.77	0.446
CEO_EducationalLevel(square)	0.005	0.060	0.08	0.937	HighTech_Industry	0.140	0.090	1.56	0.123
<i>R</i> ²	0.385				<i>R</i> ²	0.385			
<i>F</i>	3.079				<i>F</i>	3.079			
Sig. (<i>p</i> -value)	<.001				Sig. (<i>p</i> -value)	<.001			
<i>N</i>	107				<i>N</i>	107			